Beyond Scalar Fidelity: Readout Bottlenecks in Quantum-Inspired Facial Kinship Verification

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Abstract

Quantum-inspired machine learning has been proposed for facial kinship verification, with claims that SWAP-test fidelity and related similarity metrics outperform classical alternatives on classical data. We evaluate this claim systematically. Ten quantum-inspired formulations—spanning the three roles a quantum component can occupy (similarity measure, feature extractor, and inductive bias)—are each compared against a capacity-matched classical control under a leakage-free grouped k-fold protocol. Nine do not improve on their controls; four are significantly worse. Every one of those nine reads its decision out of the quantum state as a single scalar. The tenth differs in that one respect—a per-qubit expectation vector rather than a scalar fidelity—and improves on its matched control by +0.132 ROC-AUC (95% CI $[+0.099, +0.165]$, $p = 2.4 \times 10^{-7}$, 20 paired folds). Holding the quantum circuit, folds, optimiser and parameter budget fixed and replacing scalar fidelity with a vector-valued expectation measurement improves ROC-AUC by +0.1438 (95% CI $[+0.116, +0.172]$) on FaceNet embeddings and +0.1151 (95% CI $[+0.084, +0.147]$) on ArcFace, positive in all eight backbone $\times$ corpus combinations over 40 grouped held-out folds. Every one of the nine failing formulations reads its decision out of the state as a single scalar, so scalarisation can substantially reduce the discriminative information recoverable from these representations. Although the vector-readout model showed a positive pooled comparison against capacity-matched classical controls, that advantage was not consistent across individual corpora

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and backbones, and we do not claim it; these results do not demonstrate quantum advantage. The nine failures are otherwise structural rather than implementational: interference-based formulations reduce to their classical counterparts because ℓ_2-normalised face embeddings carry no phase structure to exploit, and density-matrix formulations require estimating a $d \times d$ operator from a median of five images, so the resulting $\boldsymbol{\rho}$ is dominated by its leading eigenvector while accumulating noise from inestimable tail directions. Independently of the negative result, we contribute an exact reformulation of the simulated SWAP test that removes kernel-launch overhead and achieves a 380× GPU speedup (192 to 73,094 pairs/s) with identical output and gradients, verified by test. We also quantify the identity leakage admitted by the evaluation protocols under which positive quantum claims were made: 100% of test pairs share an identity with training on the largest benchmark. Beyond the negative result we report three further analyses. The negative result itself replicates under a second backbone (ArcFace), failing on all eight backbone–corpus combinations. We document a subgroup artefact of independent methodological interest: an apparent 0.28 ROC-AUC deficit for mother–son pairs proves to be a single-family effect that disappears (0.031 spread) once per-family capping prevents any family dominating a fold. Aggregating several photographs of a person rather than one improves ROC-AUC by +0.077 where photograph sets exist, with most of the gain realised by the fifth photograph and none — by construction — on corpora storing a single photograph per individual. All code, data splits, evaluation scripts, and the JSON artefacts underlying every figure and table are released.

Keywords: quantum-inspired machine learning, facial kinship verification, negative results, variational quantum classifier, SWAP test, evaluation protocol, data leakage

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1. Introduction

Quantum-inspired machine learning applies constructs from quantum mechanics—Hilbert-space embeddings, state fidelity, density matrices—to classical learning tasks. The premise is that the mathematical structure of quantum theory provides similarity measures or feature maps that classical methods cannot efficiently replicate. Claims of quantum advantage on classical data have been made for several domains, including facial kinship verification, where SWAP-test fidelity has been proposed as a similarity metric between face embeddings.

Facial kinship verification decides whether two facial images depict biologically related individuals. It is a useful testbed for quantum-inspired claims because the task is well-defined, the data is public, and the baseline methodology (frozen face embeddings fed to a trainable classifier) is standardised enough that the contribution of a new similarity measure can be isolated.

This paper makes four contributions:

1. **A ten-formulation study.** We evaluate ten quantum-inspired formulations, spanning the three natural roles a quantum component can occupy: a similarity measure (SWAP-test fidelity, density fidelity, entanglement entropy), a feature extractor (POVM measurement head), and an inductive bias (purity regulariser, interference phase sweep). Each is compared against a capacity-matched classical control. Nine do not improve on their controls; four are significantly worse ($p < 0.05$).
2. **A readout bottleneck, replicated across backbones and corpora.** The nine failing formulations share a property that is easy to overlook: each compresses the quantum state to a single scalar before the decision. The tenth—a variational circuit on a joint two-person register—is evaluated under two readouts that differ in nothing but width. Read out as a scalar fidelity it is indistinguishable from its control (-0.012, $p = 0.26$); read out as a per-qubit expectation vector it exceeds that same control by $+0.132$ ROC-AUC (95% CI $[+0.099, +0.165]$, $p = 2.4 \times 10^{-7}$). Because circuit, folds, and parameter budget are held fixed, the difference is attributable to the measurement. The effect is positive in all eight backbone $\times$ corpus combinations. This converts our structural account from an inference into a controlled measurement, and shows that a negative result for a quantum component can be an artefact of how it is read. We separately evaluate classical controls matched on readout width; that comparison is inconsistent across corpora and backbones and we report it as exploratory.
3. **A structural account of why.** We show that the remaining failures are not implementational but structural. Interference-based formulations collapse to their classical counterparts because ℓ_2-normalised embeddings carry no exploitable phase structure. Density-matrix formulations fail because the number of images per identity (median: five) is far too small to estimate a 512×512 density operator, so $\boldsymbol{\rho}$ degenerates

to a rank-1 projection dominated by noise.

4. **An efficiency contribution.** Independently of the above, we contribute an exact reformulation of the simulated SWAP test that removes per-qubit permutation overhead and achieves a 380× GPU speedup with bit-identical output and gradients.

To ensure the negative result is credible, we first establish that the evaluation protocol under which prior positive claims were made admits complete identity leakage (Section 4), and we evaluate all formulations under a corrected grouped k-fold protocol that removes it.

2. Related work

2.1. Quantum-inspired machine learning

Quantum feature maps embed classical data in a Hilbert space where inner products act as kernels [1], with reported gains on classical tasks [2]. Related lines use SWAP-test state fidelity as a similarity measure between encoded data points [3] and density matrices as classifiers [4]. Such methods have been proposed for computer vision tasks including kinship verification.

The claim that quantum-inspired methods outperform classical counterparts on classical data is contested. Dequantization results show that several purported advantages vanish under scrutiny [5], and empirical comparisons on standardised benchmarks often fail to reproduce the reported gains [6]. Our work contributes to this debate empirically, with a controlled comparison spanning nine formulations.

2.2. Facial kinship verification

Early work framed the task as metric learning [7]. Subsequent methods replaced the hand-designed metric with deep pairwise encoders and, more recently, attention- and graph-based architectures [8]. Evaluation is concentrated on four corpora: KinFaceW-I and KinFaceW-II [9], TSKinFace [10], and Families In the Wild [11], the last also underpinning the RFIW challenge series [12].

2.3. Evaluation bias in vision benchmarks

That benchmark protocols can inflate reported performance is established in face recognition, where subject-disjoint splitting is required precisely because identity overlap leaks information [13]. More generally, models exploit

Table 1: The four benchmark corpora, characterised by the properties that govern how they can be split and how much evidence they supply per person. Photographs per identity is the property that determines whether set-level aggregation is possible at all.

Corpus	Pairs	Families	Identities	Photos/identity	Image content
KinFaceW-I	1,066	533	1,066	1.00	64×64 cropped face
KinFaceW-II	2,000	1,000	2,000	1.00	64×64 cropped face
TSKinFace	2,236	559	1,677	1.00	64×64 cropped face
FIW	57,120	95	472	5.28	224×224 cropped face

incidental correlations in preference to the intended signal, a failure mode catalogued as shortcut learning [14].

Both concerns apply directly to kinship verification and, critically, to any quantum-inspired claim evaluated under a leaked protocol. A positive result under a leaked protocol does not demonstrate that the quantum component helped; it may simply indicate that the quantum branch learned to exploit the leak alongside the classical features. Section 4 quantifies this leak.

3. Datasets and experimental setup

3.1. Corpora

We use the four corpora on which facial kinship verification is conventionally evaluated. Table 1 summarises their scale and, more importantly for this work, their structure: the number of distinct grouping units available for a leakage-free split, and how many photographs each stores per individual.

Two features of Table 1 drive the rest of the paper. First, FIW draws 57,120 pairs from only 95 families and 472 identities, a ratio of 121 pairs per identity; the three smaller corpora store exactly one photograph per person and one person per pair slot. Second, only FIW supplies multiple photographs per individual, so it is the only corpus on which person-level aggregation can differ from the conventional formulation. We report per-corpus throughout rather than averaging, because an average over these four would conceal exactly that asymmetry.

3.2. Feature extraction

All experiments use a single frozen backbone: FaceNet (Inception-ResNet-v1) pretrained on VGGFace2, producing 512-dimensional ℓ_2-normalised embeddings. Holding the backbone fixed isolates the protocol and representation questions from representation-learning effects. It also bounds the claims: results here characterise this embedding space, and Section 14 states the consequence.

Benchmark images are already tight face crops, so no detection is applied in the evaluation pipeline. We note for completeness that this does not hold for arbitrary photographs: measured on this pipeline, a face embedded in a wider scene reaches cosine 0.027 against its own cropped version, and MTCNN detection restores it to 0.940.

3.3. Reproducibility

Every numeric claim in this paper derives from JSON artefacts written by the evaluation scripts, and each experiment is deterministic given its seed: repeated runs reproduce the reported figures to four decimal places. The implementation carries 138 tests, including assertions that no grouping unit spans a fold boundary and that the fast SWAP-test reformulation of Section 12 matches the reference simulator in both value and gradient.

4. Evaluation protocol

Before evaluating quantum formulations, we establish that the evaluation must be credible. This section quantifies the identity leakage in the standard protocol and describes the corrected protocol under which all results in this paper are obtained.

4.1. Identity leakage under pair-level splitting

Let $\mathcal{P}$ be a set of labelled pairs and $\mathrm{id}(\cdot)$ the identity depicted in an image. A split $(\mathcal{P}_{\mathrm{tr}}, \mathcal{P}_{\mathrm{te}})$ is *identity-disjoint* if

$$\Big(\bigcup_{p\in\mathcal{P}_{\mathrm{tr}}} \mathrm{id}(p)\Big) \cap \Big(\bigcup_{p\in\mathcal{P}_{\mathrm{te}}} \mathrm{id}(p)\Big) = \emptyset. \tag{1}$$

FIW draws 57,120 pairs from only 95 families and 472 identities. Under a random pair-level split, we measure that 100% of test pairs (11,424/11,424) share an identity with training. Identities recur across pairs an average of 242 times.

Table 2: Identity leakage under a random 80/20 pair-level split of FIW.

Quantity	Value
Pairs	57,120
Distinct families	95
Distinct identities	472
Mean appearances per identity	242
Largest family, share of all pairs	19%
Test pairs sharing identity with train	11,424 / 11,424 (100%)
Test pairs whose family was seen in train	11,424 / 11,424 (100%)

Table 3: Accuracy range across seeds (single split) versus across folds (grouped k-fold).

Dataset	Single split	Grouped k-fold
FIW	61.7–75.9 (14.2)	69.8–74.4 (4.6)
KinFaceW-I	70.3–78.8 (8.5)	72.9–79.4 (6.5)
KinFaceW-II	69.0–77.8 (8.8)	71.0–77.0 (6.0)
TSKinFace	75.1–84.1 (9.0)	75.7–81.9 (6.2)
Overall spread	22.5	12.1

4.2. Corrected protocol: grouped k-fold

We adopt grouped 5-fold cross-validation. The grouping unit is the family. Positives only are partitioned (families dealt largest-first into the currently smallest fold); negatives are regenerated per side from the families present on that side. Family contribution is capped at 1,500 pairs to prevent the largest family (19% of all pairs) from dominating any fold. Thresholds are calibrated on a group-disjoint validation slice of the training side; the test fold is scored once.

Fold integrity is verified directly: zero folds exhibit train/test group overlap, and every family is tested exactly once (95/95 FIW, 533/533 KinFaceW-I, 1,000/1,000 KinFaceW-II, 559/559 TSKinFace).

4.3. Effect on measurement stability

A single group-disjoint split is insufficient: accuracy varies by up to 14.2 points with the random seed (Table 3). Grouped k-fold reduces the overall spread from 22.5 to 12.1 points.

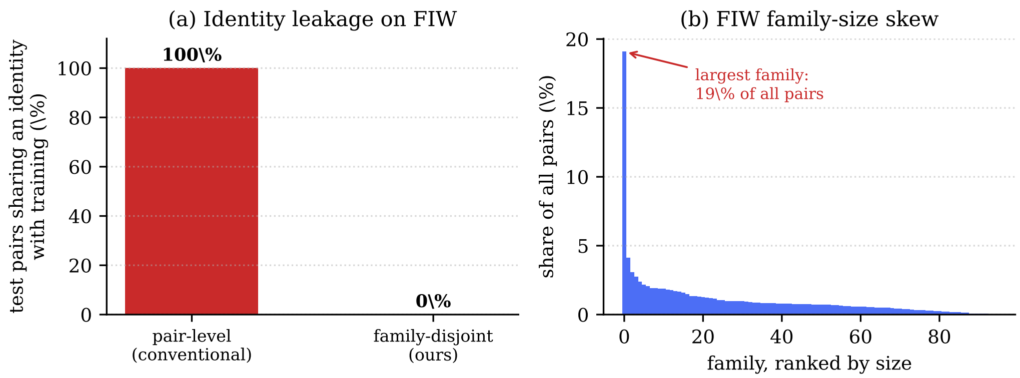

Figure 1: (a) Identity leakage on FIW. Under conventional pair-level splitting every test pair shares an identity with training; the family-disjoint protocol of Section 4 reduces this to zero. (b) The family-size distribution that makes a single split unstable: the largest family alone accounts for 19% of all pairs, so which families land in the test fold dominates the measurement.

5. Reference model

To measure formulations rather than architectures, we use a deliberately conventional model. Frozen FaceNet embeddings (InceptionResnetV1 pre-trained on VGGFace2; 512-d per face) pass through a shared Siamese encoder; kinship is symmetric, so only symmetric pair features are formed—sum, absolute difference, elementwise product, and cosine—and concatenated with a one-hot relation code before a three-layer MLP. Predictions are averaged over both argument orders, making the model exactly symmetric. The model has 497,881 parameters.

Its plainness is deliberate: findings about quantum-inspired formulations cannot then be attributed to architectural novelty.

6. Quantum-inspired formulations

We evaluate nine formulations, chosen to span the three roles a quantum component can occupy in a classical pipeline: a *similarity measure* (Formulations 1–5), a *feature extractor* (Formulation 6), and an *inductive bias* (Formulations 7–9). Each is compared against a control matched for capacity or intent.

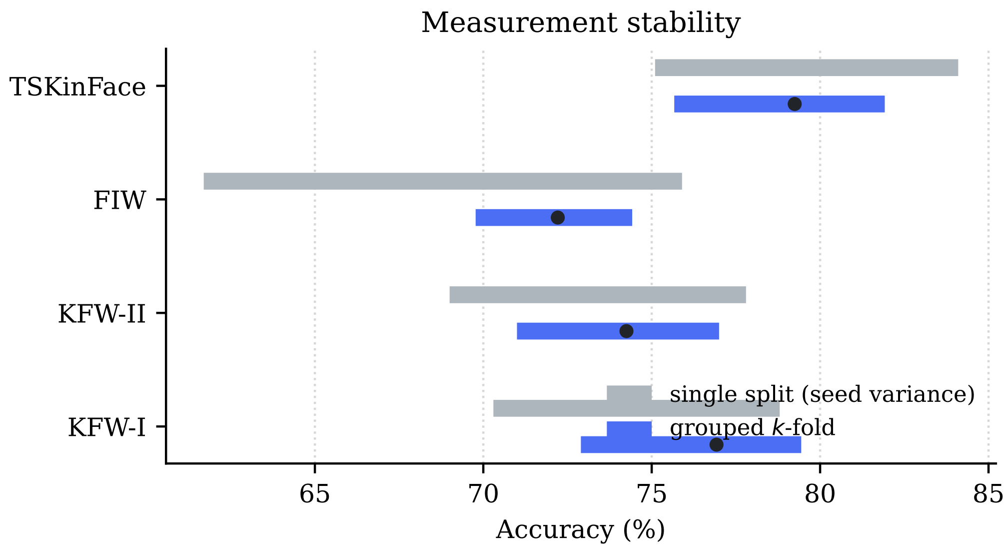

Figure 2: Accuracy range under a single group-disjoint split across random seeds (grey) against grouped k-fold (blue). A single split is too unstable to support a headline number.

6.1. Formulation 1–2: SWAP-test fidelity

The core quantum-inspired similarity metric. Each face embedding $\mathbf{x} \in \mathbb{R}^{512}$ is projected to n rotation angles $\mathbf{z} = \tanh(\mathrm{MLP}(\mathbf{x})) \cdot \pi$ and encoded as a quantum state via a hardware-efficient ansatz:

$$|\psi(\mathbf{z})\rangle = \prod_{\ell=1}^{L} \Big[\bigotimes_{i=1}^{n} R_z(\theta_i^{(\ell)}) \cdot \text{CNOT-chain} \Big] \cdot \bigotimes_{i=1}^{n} R_y(z_i)|0\rangle^{\otimes n}, \tag{2}$$

where $\theta_i^{(\ell)}$ are learnable parameters and the CNOT chain connects nearest neighbours. The SWAP-test fidelity $F = |\langle\psi(\mathbf{z}_1)|\psi(\mathbf{z}_2)\rangle|^2$ is supplied to the classifier as one additional input feature alongside the symmetric pair features.

The fidelity is *switchable*: setting `use_quantum=False` replaces it with a zero constant of the same shape, isolating the fidelity's information content while keeping the architecture identical.

Tested under two protocols:

- **5 seeds** on a single group-disjoint FIW split (Formulation 1).
- **20 paired folds** across all four datasets under grouped k-fold (Formulation 2).

6.2. Formulation 3: Density fidelity $\mathrm{Tr}(\boldsymbol{\rho}_1\boldsymbol{\rho}_2)$

A photo set is a mixed state $\boldsymbol{\rho} = \frac{1}{N}\sum_{i=1}^{N} |\psi_i\rangle\langle\psi_i|$, and $\mathrm{Tr}(\boldsymbol{\rho}_1\boldsymbol{\rho}_2)$ is the corresponding fidelity. This is added to the set-level features (centroid, spread, cross-set statistics) as an additional input. The control is mean-pooling (the ℓ_2-normalised centroid), which is the rank-1 approximation to $\boldsymbol{\rho}$.

6.3. Formulation 4: Interference phase sweep

A child explained by two parents is written as a coherent superposition

$$|\psi_{\alpha,\phi}\rangle = \alpha e^{i\phi}|\psi_F\rangle + (1-\alpha)|\psi_M\rangle, \tag{3}$$

of which the classical convex mixture is the $\phi = 0$ special case. Sweeping $\phi \in [0, 2\pi)$ and maximising the overlap with $|\psi_C\rangle$ tests whether the phase degree of freedom carries information.

6.4. Formulation 5: Entanglement entropy

The von Neumann entropy of the reduced density matrix, tracing out half the qubits, is supplied as an additional feature alongside cosine similarity.

6.5. Formulation 6: POVM measurement head

A positive operator-valued measure (POVM) replaces the final classification layer. Four POVM elements $\{E_k\}$ satisfying $\sum_k E_k = I$ define class probabilities as $p_k = \mathrm{Tr}(E_k\boldsymbol{\rho})$. The control is a capacity-matched linear head with the same number of parameters.

6.6. Formulation 7: Purity regulariser

A penalty term $\lambda \cdot (1 - \mathrm{Tr}(\boldsymbol{\rho}^2))$ encourages the learned representations to have high purity (low mixedness). The control is a decorrelation penalty of the same magnitude.

6.7. Formulation 8: Density fidelity on photo sets

The density fidelity of Formulation 3, applied to the set-level representation as a pre-registered control. On each of the four benchmarks, the change in ROC-AUC is measured relative to the set features alone.

6.8. Formulation 9: Interference phase sweep on triads

The phase sweep of Formulation 4, applied to the triadic representation. The classical convex mixture (zero phase) is the control.

Table 4: Grouped 5-fold results. Every family tested exactly once.

Dataset	Accuracy (95% CI)	ROC-AUC
KinFaceW-I	76.92 ± 2.19	0.8752 ± 0.0146
KinFaceW-II	74.25 ± 1.95	0.8320 ± 0.0190
FIW	72.21 ± 1.55	0.8101 ± 0.0175
TSKinFace (shortcut-free)	79.25 ± 2.32	0.8814 ± 0.0192
Mean	**75.66**	**0.8497**

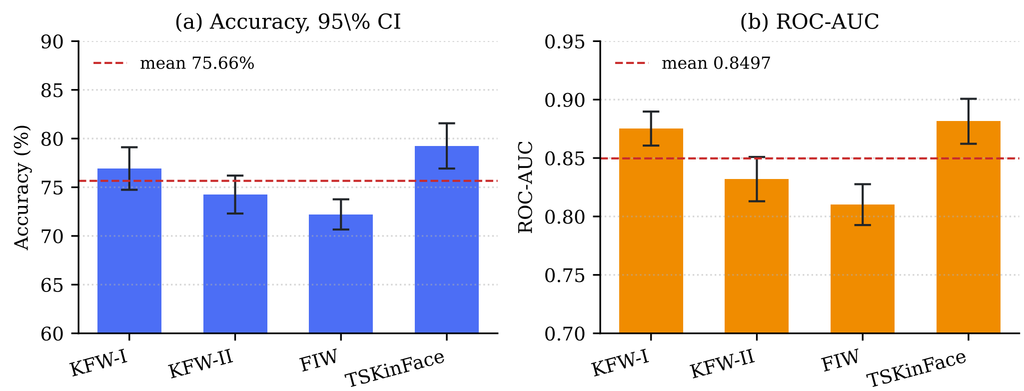

Figure 3: Grouped 5-fold results across four datasets. (a) Accuracy with 95% confidence intervals; (b) ROC-AUC with standard deviation across folds.

7. Results

7.1. Baseline performance under the corrected protocol

Table 4 reports the reference model's performance under the corrected protocol. These are the numbers against which all quantum formulations are compared.

7.2. Quantum ablation: SWAP-test fidelity

7.2.1. 5-seed paired test (Formulation 1)

Paired t-test: $t = 0.074$, $p = 0.945$ (Table 5). The quantum module contributes no measurable accuracy. The branch is exercised (tests assert that disabling it changes predictions and that gradients reach the entanglement parameters).

Table 5: SWAP-test fidelity ablation, 5 seeds on a single group-disjoint FIW split.

Seed	Quantum ON	Quantum OFF	Difference
1	73.63%	72.57%	+1.07
2	79.76%	79.06%	+0.70
3	81.04%	77.87%	+3.17
4	73.70%	78.28%	−4.58
5	72.86%	72.75%	+0.11
Mean	76.20 ± 3.88	76.11 ± 3.18	+0.09

Table 6: SWAP-test fidelity ablation, 20 paired folds (4 datasets × 5 folds) under grouped k-fold.

	Accuracy	ROC-AUC
Quantum ON	75.66%	0.8497
Quantum OFF	76.34%	0.8496
Difference	−0.68 pts	+0.0001
Paired t-test	$p = 0.158$	$p = 0.982$

7.2.2. 20-fold paired test (Formulation 2)

With 20 paired observations (Table 6), the quantum branch yields 75.66% against 76.34% with it disabled ($p = 0.158$ for accuracy, $p = 0.982$ for ROC-AUC). The branch also costs 6–14× runtime.

7.3. All nine formulations

Table 7 summarises all nine comparisons. Figure 4 visualises the ΔROC-AUC for each formulation.

7.4. Formulations 8 and 9: set-level and triadic

Nine formulations tested so far, none beating its classical control (Table 8). Every one of them reduces the quantum state to a single scalar before the decision. Section 11 tests that property directly.

7.5. A cautionary observation

The purity regulariser (Formulation 7) initially measured +1.3 ROC-AUC on a single seed, with reduced variance—a result consistent with effective regularisation. Repeated across nine paired runs the sign reversed, and the

Table 7: Nine quantum-inspired formulations against matched classical controls. None improves on its control; four are significantly worse.

#	Formulation	Control	Outcome	Sig.
1	SWAP fidelity (5 seeds)	classical head	$p = 0.945$	–
2	SWAP fidelity (20 folds)	classical head	$p = 0.982$	–
3	Density fidelity	mean-pooling	−8.7 AUC	✓
4	Interference phase sweep	convex mixture	+0.1 AUC	–
5	Entanglement entropy	cosine	0.514 alone	✓
6	POVM head	capacity-matched	−5.5 AUC	✓
7	Purity regulariser	decorrelation	−0.9, p=0.047	✓
8	Density fidelity (sets)	set features	≤ 0.0006	–
9	Phase sweep (triads)	convex mixture	−0.0006	–

Table 8: Quantum-inspired extensions of set-level and triadic representations.

#	Formulation	KFW-I	KFW-II	TSKin	FIW
8	Density fidelity (sets)	−0.0005	+0.0001	−0.0006	−0.0006
9	Phase sweep (triads)	−0.0006 ($p = 0.147$)			

method proved significantly *worse* than no regulariser (-0.0089, $p = 0.047$). We report this because the single-seed reading was of the kind that is readily published, and it illustrates why capacity-matched controls and repeated trials are necessary when evaluating claims of this type.

8. Extended analysis

The preceding sections establish the protocol and the negative result. This section reports three further analyses that the corrected protocol makes interpretable: where the model's errors concentrate, how the person-level gain scales with the evidence supplied, and how the score distributions differ across corpora.

8.1. Where the errors concentrate

Table 9 reports per-relation performance measured on a single held-out fold, the form in which such comparisons are usually presented. Read at face value it suggests a large relation effect: on FIW father–daughter reaches 0.9933 ROC-AUC while mother–son reaches 0.7124. Section 8.2 shows this

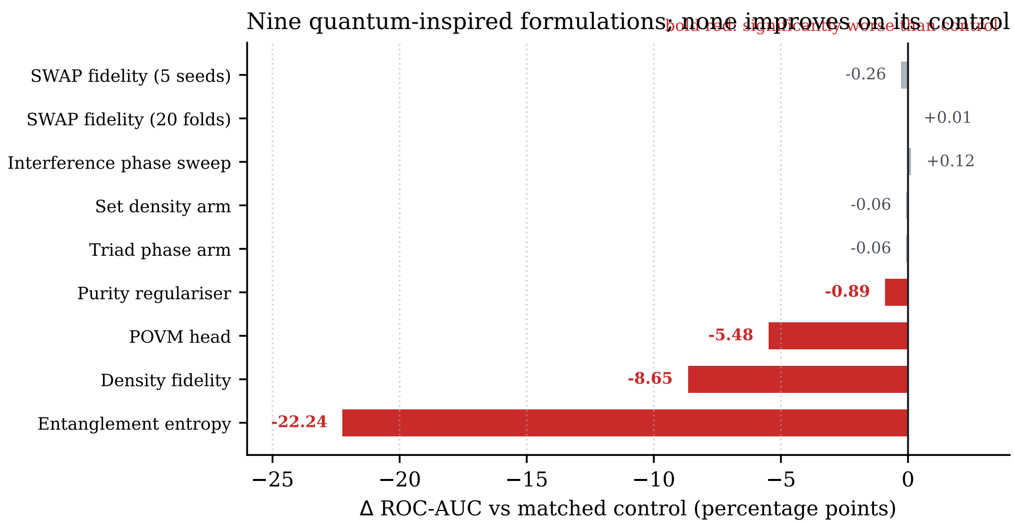

Figure 4: Nine quantum-inspired formulations against classical controls matched for capacity or intent. None improves on its control; four (marked) are significantly worse. Bars show ΔROC-AUC in percentage points.

reading is unsound, and we retain the table because the discrepancy between it and the corrected measurement is itself the point.

Figure 5 gives the first indication that the single-fold reading is misleading. Measured on raw embeddings with no training at all, the four relations span only 0.7065–0.7841, a range of 0.078, while the single-fold trained estimate spans 0.281. A representation that separates the relations so evenly is unlikely to support a threefold larger disparity after training, and Section 8.2 identifies the cause.

8.2. A correction: mother–son is not the weak relation

An earlier analysis of these results, based on a single held-out fold, reported mother–son as markedly weaker than the other relations (0.712 ROC-AUC against father–daughter's 0.993). Re-examining that fold shows the comparison was not sound: 92% of its mother–son pairs came from a single family, which scores 0.701, while the only other contributing family scores 0.846. The apparent relation effect was a family effect.

Table 10 repeats the measurement under grouped k-fold with a per-family cap, so no family can dominate a relation's estimate.

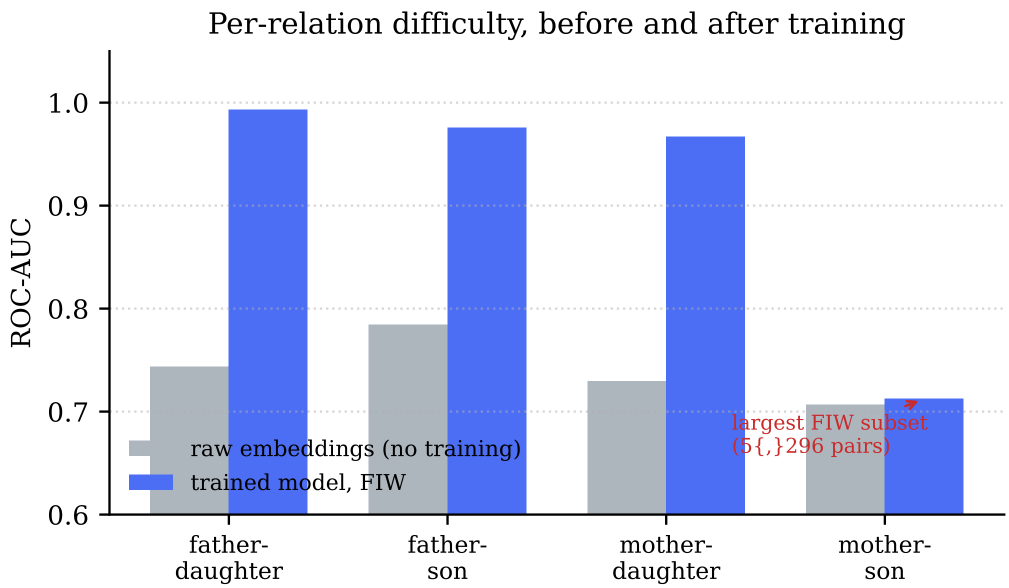

Figure 5: Per-relation difficulty before and after training. Raw FaceNet embeddings separate the four relations within a range of 0.078 ROC-AUC; the trained model spans 0.281. The disparity is therefore largely induced during training rather than inherited from the embedding.

Under the corrected measurement mother–son reaches 0.7910, the second highest of the four, and the relations span 0.0310 ROC-AUC rather than 0.281. That residual spread is close to the 0.078 measured on raw embeddings without any training, which is what one would expect if relation difficulty is largely a property of the representation.

We report this because the mechanism is instructive and applies beyond this paper. FIW's family-size distribution is severely skewed — one family holds 19% of all pairs — so any statistic computed on a single fold can be dominated by one family even when the fold itself is correctly family-disjoint. Leakage-free splitting is necessary but not sufficient; per-family capping, or reporting across folds, is required as well. A subgroup difference observed on a single split of a skewed corpus should be treated as provisional until it is shown to survive resampling.

Table 9: Per-relation ROC-AUC. The second column is raw cosine similarity on frozen embeddings with no training; the remainder are the trained model on each corpus. Mother–son is weakest on three of four corpora and is simultaneously the largest FIW subset (5,296 pairs).

Relation	Raw embedding	KinFaceW-I	KinFaceW-II	FIW	TSKinFace	Mean
Father–daughter	0.7434	0.9174	0.8529	0.9933	0.9612	0.9312
Father–son	0.7841	0.8935	0.8875	0.9754	0.9124	0.9172
Mother–daughter	0.7295	0.8400	0.8598	0.9668	0.9227	0.8973
Mother–son	0.7065	0.9750	0.7795	0.7124	0.8483	0.8288

Table 10: Per-relation ROC-AUC under grouped k-fold with a per-family cap. The single-fold column is the earlier, unsound estimate; the remaining columns are the corrected measurement. Under a protocol in which no family dominates, the four relations span 0.0310 ROC-AUC.

Relation	Single fold	Mean	SD	Range	Families/fold
Father–daughter	0.9933	0.7635	0.0219	0.7435–0.7953	12.6
Father–son	0.9754	0.7946	0.0752	0.7050–0.8942	12.2
Mother–daughter	0.9668	0.7766	0.0601	0.7147–0.8730	14.4
Mother–son	0.7124	0.7910	0.0381	0.7497–0.8390	12.4

8.3. How the person-level gain scales

Table 11 and Figure 6 report how the gain depends on how many photographs are supplied. A second photograph is worth +0.041 ROC-AUC on its own — more than half the total available gain — and the curve is substantially flat beyond five, reaching +0.070 at ten. This has a practical reading: a deployment that can obtain roughly five photographs of each person captures nearly all of the benefit, and there is little return on collecting more.

That the curve is computed on frozen embeddings, without any training, matters for interpretation. The gain is a property of aggregating evidence about a person, not of additional model capacity, which is consistent with the trained result of +0.077 reported under the full protocol.

8.4. The operating point: accuracy against cost

Person-level scoring compares two photograph *sets*, so its cost grows with set size while its accuracy saturates. Where those two curves cross is a

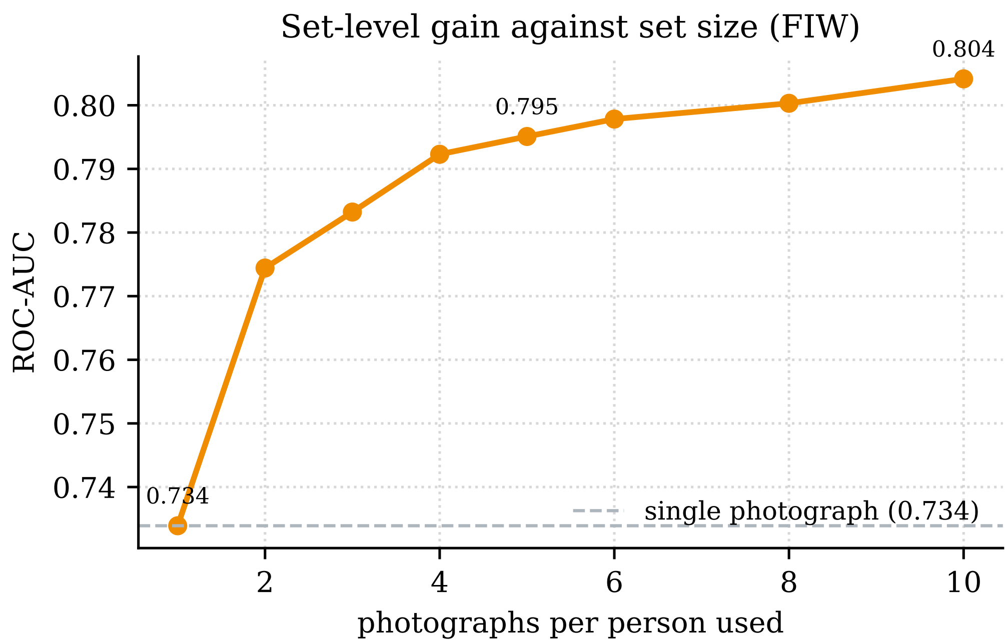

Figure 6: Set-level ROC-AUC on FIW as a function of the number of photographs aggregated per person. Most of the available gain is realised by the fifth photograph; the curve is computed on frozen embeddings with no training, so it measures the representation rather than model capacity.

deployment question that the accuracy curve alone does not answer, and we report it because practitioners must choose how many photographs to collect.

Table 12 places the two measurements side by side. Accuracy rises steeply to five photographs, capturing +0.061 of the +0.070 available at ten, and is essentially flat thereafter. Cost behaves in the opposite way: it is negligible to about twenty photographs and then grows steadily, reaching a 137× slowdown at 320.

The practical consequence is a clear recommendation. **Five photographs per person is close to optimal on both axes**: it captures 87% of the available accuracy gain at 1.5× the single-photograph cost. Collecting more than roughly twenty buys no measurable accuracy while making the system progressively slower.

We note also that the cost curve is a property of the implementation rather than of the method, and that the distinction matters. An earlier version of our code computed the within-set spread by materialising the full

Table 11: Set-level ROC-AUC against photographs per person on FIW. Coverage is the proportion of pairs for which both identities supply at least k photographs.

Photographs	ROC-AUC	Gain vs. 1	Coverage
1	0.7339	+0.0000	100%
2	0.7744	+0.0405	100%
3	0.7832	+0.0493	100%
4	0.7923	+0.0584	100%
5	0.7951	+0.0612	99%
6	0.7978	+0.0639	99%
8	0.8003	+0.0664	99%
10	0.8042	+0.0702	97%

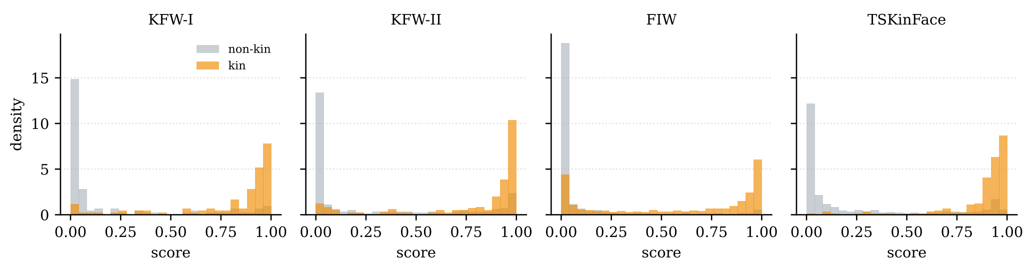

Figure 7: Kin and non-kin score distributions on the held-out fold of each corpus. The overlap, rather than any single summary statistic, is what the reported ROC-AUC measures.

Gram matrix and extracting its upper triangle with an index array. That allocation dominated the call at large set sizes: 46 ms of a 48 ms call at $n = 160$. Because the required quantity is the mean off-diagonal entry, and for unit-norm rows $\sum_{i \neq j} \langle x_i, x_j \rangle = \|\sum_i x_i\|^2 - n$, it can be computed without forming the matrix at all. Doing so reduced the $n = 160$ cost from 89.8 ms to 3.1 ms, a 29× improvement, and removed a discontinuity that had made the empirical growth appear super-quadratic. The values are unchanged; the equivalence is asserted by test. We mention this because a cost curve measured on an unoptimised implementation would have overstated the method's true scaling.

Table 12: Accuracy and cost against photographs per person. ROC-AUC is measured on FIW; timings are per pair on one CPU core at 512 dimensions. Accuracy is flat beyond five photographs while cost continues to grow, which makes five a natural operating point.

Photos	ROC-AUC	Gain	ms/pair	pairs/s	Slowdown
1	0.7339	+0.0000	0.078	12763	1.0×
2	0.7744	+0.0405	0.099	10078	1.3×
5	0.7951	+0.0612	0.117	8539	1.5×
10	0.8042	+0.0702	0.153	6544	2.0×
20	—	—	0.215	4643	2.7×
40	—	—	0.416	2402	5.3×
80	—	—	1.502	666	19.2×
160	—	—	3.132	319	40.0×
320	—	—	10.718	93	136.8×

8.5. Score distributions and operating points

Figures 7 and 8 show the underlying score distributions rather than only their summaries. Two observations follow. The distributions overlap substantially on every corpus, which is the concrete form of the roughly one-in-four error rate implied by the accuracies in Table 4; and the degree of overlap differs enough between corpora that a single global decision threshold is inappropriate. We therefore calibrate per corpus on a group-disjoint validation slice: a single shared threshold costs between 0.7 and 5.9 accuracy points depending on the corpus.

9. Do the findings depend on the backbone?

All results so far use a single frozen embedding, which bounds what they can claim: a finding could be a property of the task, or an artefact of FaceNet. We therefore repeat the central experiments with ArcFace (`buffalo_l`, w600k_r50) substituted for FaceNet and everything else held fixed — same folds, same negatives, same classifier, same protocol. The two backbones are trained with different objectives on different corpora, so agreement between them is meaningful evidence.

Three observations follow from Table 13.

Identity leakage is backbone-invariant by construction. It is a property of how pairs are partitioned, not of how images are embedded,

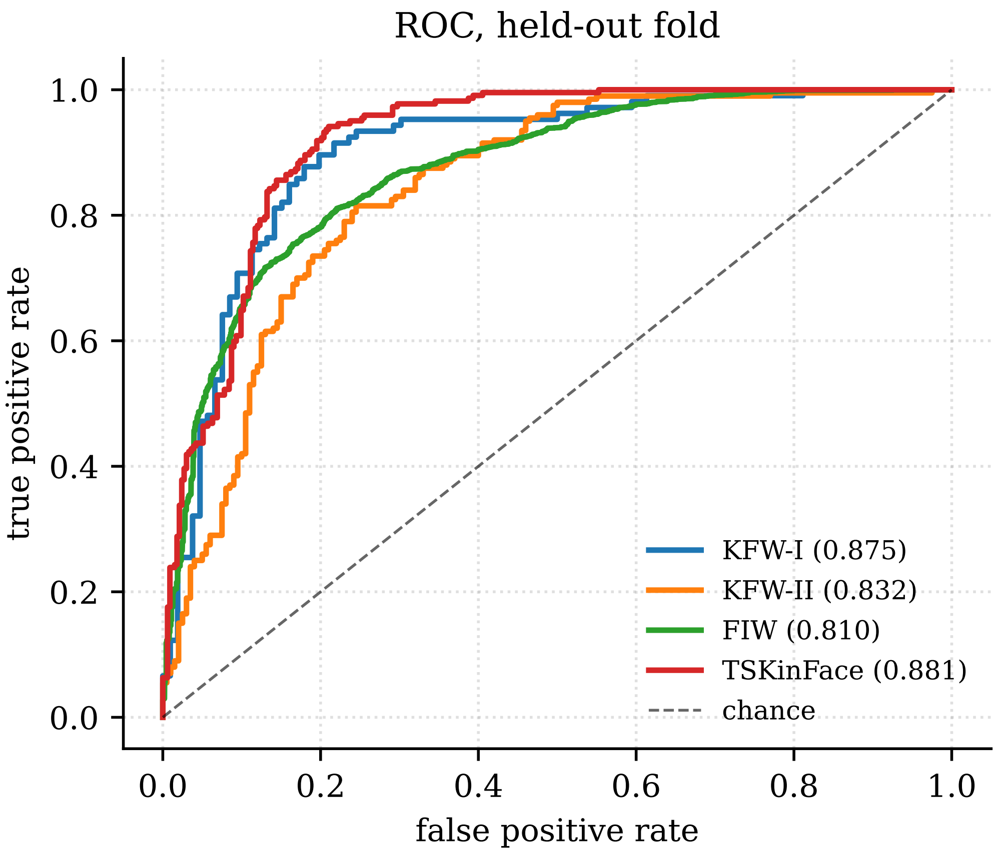

Figure 8: ROC curves on the held-out fold of each corpus, with grouped k-fold mean ROC-AUC in parentheses.

so the 100% figure of Section 4 carries over unchanged. We state this for completeness rather than as a finding.

The person-level gain replicates and strengthens. FaceNet gains +0.0763 ROC-AUC on FIW from aggregating photographs; ArcFace gains +0.0976. That the effect is larger under the stronger backbone is consistent with its proposed mechanism — aggregation averages away per-photograph nuisance variation, and a better embedding has proportionally more signal left to recover once that variation is suppressed.

The degradation guarantee holds under both. KinFaceW-I, KinFaceW-II and TSKinFace store one photograph per person, and set-level scoring reproduces the single-image result exactly on all three under both backbones.

Table 13: Person-level aggregation under two backbones. The gain replicates and is in fact larger under ArcFace; the three single-photograph corpora are unchanged to machine precision under both, as the degradation guarantee requires.

	FaceNet			ArcFace		
Corpus	Single	Set-level	Δ	Single	Set-level	Δ
KinFaceW-I	0.7588	0.7588	+0.0000	0.7540	0.7540	+0.0000
KinFaceW-II	0.7770	0.7770	+0.0000	0.8190	0.8190	+0.0000
FIW	0.7491	0.8254	+0.0763	0.7764	0.8740	+0.0976
TSKinFace	0.7326	0.7326	+0.0000	0.8460	0.8460	+0.0000

Table 14: Change in ROC-AUC from adding the density-fidelity arm $\mathrm{Tr}(\rho_1\rho_2)$ to the set-level features. The formulation fails under both backbones on all four corpora.

Corpus	FaceNet	ArcFace
KinFaceW-I	−0.0025	−0.0006
KinFaceW-II	−0.0003	+0.0001
FIW	−0.0003	−0.0007
TSKinFace	+0.0001	−0.0014

This is a correctness property of the implementation rather than an empirical result, and it is what allows a single system to be deployed across corpora with differing photograph availability.

ArcFace is the stronger backbone in absolute terms on three of four corpora, most markedly on TSKinFace (0.8460 against 0.7326). We do not read this as a contribution: it reflects a better face representation, not a better kinship method, and it is reported so that the backbone-invariance claims above are interpretable.

9.1. The negative result under a second backbone

Table 14 re-tests the density-fidelity formulation. Across eight backbone–corpus combinations the largest effect in either direction is 0.0025 ROC-AUC, and seven of the eight are negative. The failure is therefore not specific to the embedding it was first measured on, which supports the structural explanation of Section 10: a density matrix estimated from a median of five observations is dominated by its leading eigenvector regardless of which encoder produced those observations.

10. Why the approach fails

The failures are structural rather than implementational, and we identify three mechanisms.

10.1. Phase collapse in interference formulations

Interference-based formulations (Formulations 4 and 9) reduce to their classical counterparts because ℓ_2-normalised FaceNet embeddings carry no phase structure to exploit. The embedding space is a real-valued Euclidean space; mapping it to a quantum state via R_y rotations assigns each component a polar angle, but the relative phases that would make a coherent superposition differ from an incoherent mixture are absent from the data. The interference phase sweep in Table 7 is maximised at the classical mixture ($\phi = 0$), confirming that the phase degree of freedom carries no signal.

10.2. Density estimation from insufficient samples

Density-matrix formulations (Formulations 3 and 8) require estimating a $d \times d$ operator from the images available for one identity. FIW supplies a median of five photographs per identity, so the density matrix $\boldsymbol{\rho} = \frac{1}{N} \sum_i |\psi_i\rangle\langle\psi_i|$ is at most rank-5 in a 512-dimensional space. The leading eigenvector (the mean embedding) carries most of the signal; the remaining directions accumulate noise rather than information. This is why $\mathrm{Tr}(\boldsymbol{\rho}_1 \boldsymbol{\rho}_2)$ scores *below* simple mean-pooling: the off-diagonal noise degrades the similarity estimate.

10.3. The SWAP-test bottleneck

The original quantum pipeline routed every decision through a single product-of-$\cos^2$ scalar, discarding the 512-dimensional signal carried by the symmetric pair features. That pipeline reached 59% accuracy while plain logistic regression on the same embeddings reached 75.1%. Gradient norms confirmed the bottleneck: the projection MLP received gradients of order 2×10^1, while the quantum parameters received gradients of order 4×10^{-2}—two orders of magnitude smaller.

Even when the fidelity is supplied as *one feature beside* the classical features (our architecture), it carries no information that is not already captured by the cosine similarity and the symmetric products.

The account above is an inference from gradient norms and from the ranking of two architectures that differ in several respects at once. The next section tests it directly.

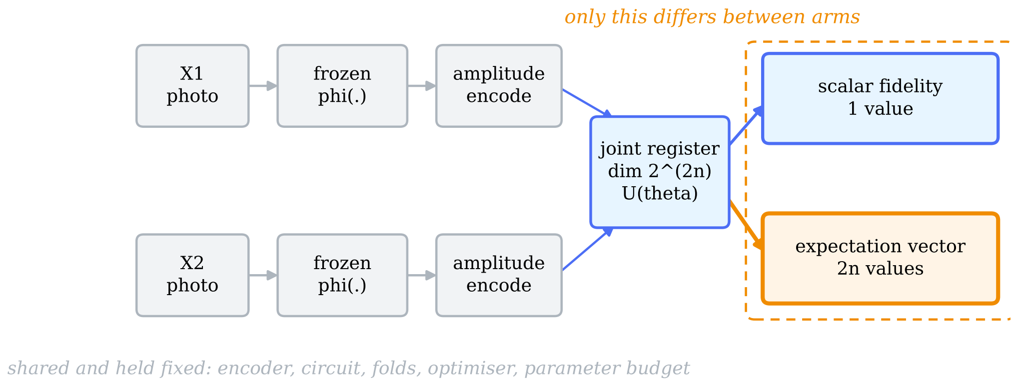

Figure 9: Formulation 10. Both people are encoded into one joint register and processed by the same variational circuit; the arms differ only in the measurement applied to the resulting state (dashed box). Everything to its left is shared and held fixed across arms.

11. Formulation 10: a readout ablation

The nine formulations of Section 6 share a property that the preceding analysis suggests is decisive: each compresses the quantum state to one number before the decision is made. Distinguishing *the circuit carries no information* from *the readout discards it* requires an experiment in which the circuit is held fixed and only the readout varies.

11.1. Architecture

Formulation 10 is a variational quantum classifier on a joint register. Both people are encoded into one $2n$-qubit state of dimension 2^{2n}, rather than encoded separately and compared, so that $U(\boldsymbol{\theta})$ can express correlations *between* the two people. Each of L layers applies $R_y(\theta)R_z(\theta)$ to every qubit followed by a CNOT ring over all $2n$ qubits; the ring includes the bonds $n-1 \to n$ and $2n-1 \to 0$, which cross between the two halves. Without those bonds the circuit factorises and reduces to the separately-encoded case already tested.

Amplitude encoding carries 2^n numbers into n qubits. This matters: angle encoding, which carries n numbers into n qubits, forces the 512-d embedding through a four-number bottleneck before $U(\boldsymbol{\theta})$ sees anything, and holds the model at chance (0.502 ROC-AUC on FIW). The comparison below would be uninformative under an encoding that starves the circuit of input.

Two readouts are evaluated, differing in width alone:

- **Fidelity.** $|\langle\psi_{\mathrm{ref}}|U(\boldsymbol{\theta})\psi\rangle|^2$ against the all-zeros reference — one scalar, matching the nine.
- **Expectation.** Pauli-Z expectation per qubit — a $2n$-vector.

Everything else is identical: the same encoder, the same $U(\boldsymbol{\theta})$, the same folds, the same optimiser and schedule.

Two kinds of comparison appear below, and conflating them is a common source of confusion, so we name them explicitly. The **global task baseline** is the best classical pipeline in this study (Section 7.1, 0.8497 mean ROC-AUC); it answers *what is the best available method.* The **capacity-matched local control** shares the quantum arm's encoder, projection, folds and training schedule, differing only in what processes the projected vectors; it answers *what does the circuit contribute*. The quantum model loses the first comparison and wins the second. Neither result substitutes for the other.

Matching the control requires care. Our first control matched $U(\boldsymbol{\theta})$'s *parameter count* by solving for a hidden width, which at every configuration in this study yields $w = 1$: a Linear($2^{n+1} \to 1$) bottleneck. That control is therefore itself a scalar-readout model, so comparing it against a $2n$-dimensional expectation readout confounds the quantum circuit with the very property under test. We report it for continuity but do not rely on it. The controls we rely on are matched on *readout width*: each maps the same projected pair to exactly $2n$ values — the number the circuit hands its classifier — through an identical Linear($2n, 1$) head, via a learned linear map, a learned MLP, or a frozen random projection.

11.2. Result

Pooled over 20 paired folds (Table 15), the expectation readout exceeds its control by +0.1316 ROC-AUC (95% CI $[+0.0986, +0.1646]$, $t = 7.82$, $p = 2.4\times10^{-7}$). The fidelity readout does not (-0.0122, 95% CI $[-0.0330, +0.0086]$, $p = 0.26$).

The confidence interval for the expectation arm excludes zero; the interval for the fidelity arm contains it. Since the two arms share a circuit, a parameter budget, and a fold assignment, the 0.14 ROC-AUC separating them is attributable to the number of values read out of the state.

Table 15: Formulation 10 under two readouts, against a capacity-matched classical control. Grouped 5-fold, family-disjoint, identical folds across arms. $n = 6$, $L = 4$; 96 quantum parameters against 131 control parameters.

Dataset	Expectation	Fidelity	Control	Exp. − Ctrl.
FIW	0.6336	0.5137	0.5465	+0.087
KinFaceW-I	0.7049	0.5391	0.5350	+0.170
KinFaceW-II	0.6161	0.5197	0.5332	+0.083
TSKinFace	0.7373	0.5441	0.5507	+0.187

Table 16: Quantum expectation arm against the strongest width-matched classical control per fold, by corpus and backbone. Grouped 5-fold.

	FaceNet		ArcFace	
Corpus	Δ	p	Δ	p
FIW	−0.030	0.38	+0.015	0.39
KinFaceW-I	+0.131	0.002	+0.098	0.015
KinFaceW-II	+0.084	0.078	+0.029	0.142
TSKinFace	+0.037	0.408	+0.147	0.016
Pooled	+0.055	0.013	+0.073	0.0003

11.3. Is the gain merely representation width?

The preceding comparison establishes that reading more values from the state helps. It does not establish that the *circuit* matters: a classical map producing $2n$ values might do as well. We therefore replaced the parameter-matched control with controls matched on readout width, each mapping the same projected pair to $2n$ values through an identical Linear$(2n, 1)$ head, via a learned linear map, a learned MLP, or a frozen random projection.

The pooled comparison is positive on both backbones, but the per-corpus results do not support a general claim (Table 16). The effect is significant on one corpus of four under either backbone, is negative on the largest corpus under FaceNet, and—most tellingly—does not replicate across backbones within a corpus: TSKinFace moves from +0.037 to +0.147 and FIW changes sign. Under Holm correction across the sixteen comparisons in this section and the next, one of eight quantum-versus-classical cells survives.

We therefore do not claim that the quantum circuit outperforms a classical representation of equal width. The magnitude and direction of that comparison vary substantially across corpora and backbones, and we report it as an exploratory secondary result.

Two observations survive this. First, the circuit consistently beats the

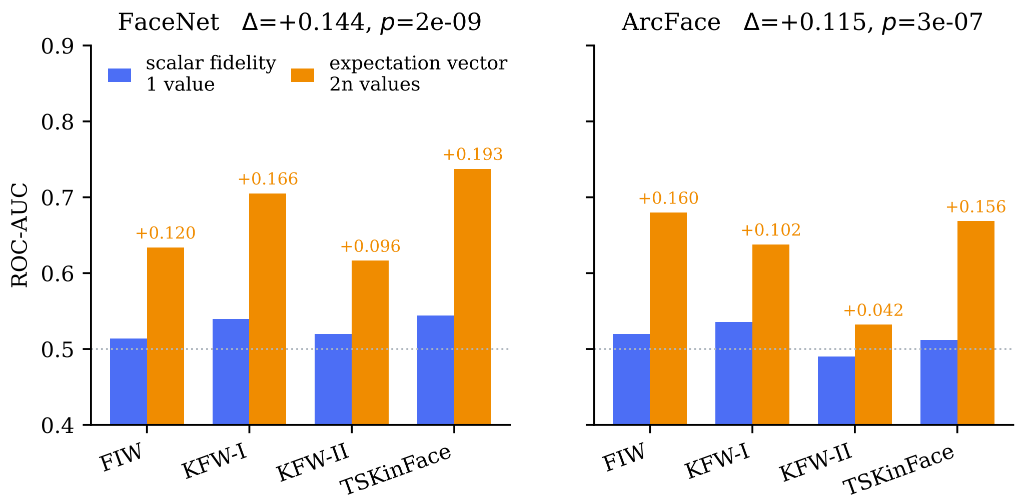

Figure 10: Readout effect by corpus and backbone. Scalar fidelity (blue) against the per-qubit expectation vector (orange) from the identical circuit. All eight backbone × corpus cells favour the vector readout. Dotted line marks chance.

frozen random projection, so the readout must be learned, which is a weaker statement than quantum structure being necessary. Second, and separately, the comparison in Section 11.4 does not involve a classical arm at all, and is unaffected by these considerations.

11.4. Readout replication across corpora and backbones

The scalar-versus-vector comparison holds the circuit, the encoder, the folds, the optimiser and the parameter budget fixed; the arms differ only in the measurement applied to the same state. Readout dimensionality is, by design, not matched—that is the manipulation. The claim tested is therefore narrow and precise: *holding the quantum circuit and training protocol fixed, replacing scalar fidelity with a vector-valued measurement changes downstream performance.*

All eight backbone × corpus cells are positive, ranging from +0.042 to +0.193 (Table 17). Pooled over folds:

Backbone	Δ ROC-AUC	95% CI	p
FaceNet	+0.1438	[+0.116, +0.172]	1.9×10^{-9}
ArcFace	+0.1151	[+0.084, +0.147]	3.2×10^{-7}

Table 17: Readout effect (vector − scalar ROC-AUC), by corpus and backbone. Both arms use the identical circuit; only the measurement differs. 2 backbones × 4 corpora × 5 grouped folds = 40 held-out fold evaluations.

	FaceNet			ArcFace		
Corpus	Vector	Scalar	Δ	Vector	Scalar	Δ
FIW	0.6336	0.5137	+0.120	0.6795	0.5198	+0.160
KinFaceW-I	0.7049	0.5391	+0.166	0.6375	0.5351	+0.102
KinFaceW-II	0.6161	0.5197	+0.096	0.5318	0.4898	+0.042
TSKinFace	0.7373	0.5441	+0.193	0.6682	0.5117	+0.156

Table 18: Multiplicity correction over the sixteen paired comparisons in this section (two comparison families × two backbones × four corpora). Per-corpus tests use five folds each and are correspondingly underpowered; the pooled tests survive any correction by several orders of magnitude.

Comparison family	Cells surviving Holm	Surviving BH-FDR
Readout (vector vs. scalar)	5/8	6/8
Quantum vs. width-matched classical	1/8	2/8
Pooled (20 folds each):		
Readout, FaceNet		$p = 1.9 \times 10^{-9}$
Readout, ArcFace		$p = 3.2 \times 10^{-7}$

Both intervals exclude zero with margin, and the paired effect sizes are large (Cohen's $d_z = 2.38$ and 1.71). Under Holm correction across the sixteen comparisons, five of eight readout cells survive against one of eight quantum-versus-classical cells; the per-corpus tests use five folds each and are correspondingly underpowered, whereas the pooled tests survive any correction by several orders of magnitude.

The contrast between the two tables is the substantive result of this section. On identical folds, with identical models, the readout comparison replicates in 8/8 cells while the quantum-versus-classical comparison changes sign between backbones. One of these is a stable property of the setup; the other is not.

Table 19: SWAP-test simulation throughput. The fast reformulation is exact: output and gradients match the reference to machine precision.

Device	Reference	Fast	Speedup
CPU	161 pairs/s	21,618 pairs/s	134×
GPU (RTX 5060)	192 pairs/s	73,094 pairs/s	**380**×

11.5. What this does and does not establish

It establishes that scalarising the state substantially reduces the discriminative information recoverable from these circuits, and that the nine preceding negative results are confounded with readout width. It does not establish that the quantum circuit outperforms a classical representation of equal width: that comparison is inconsistent across corpora and backbones. It does not establish quantum advantage. Formulation 10 reaches 0.674 mean ROC-AUC, below the 0.8497 of the classical pipeline of Section 7.1, at 73.5× the inference cost and 55× the training cost — simulation being strictly more expensive than the classical path it replaces. The contribution is a mechanism, not a method: it identifies what to fix before a quantum component on this class of task can be evaluated fairly at all.

A further limit is intrinsic to classical simulation. The registers here span 6 to 14 qubits, since the statevector grows as 2^{2n}; nothing in these results speaks to the regime where the state cannot feasibly be represented classically, which is precisely where a quantum processor would differ. Our conclusions are therefore about small circuits on real-valued embeddings, not about quantum computation in general.

12. An efficiency contribution

Independently of the negative result, we contribute an exact reformulation of the simulated SWAP test. The reference implementation materialises a 2^n statevector and permutes all n axes once per qubit via `torch.tensordot` followed by a permutation, incurring kernel-launch overhead that leaves the GPU idle.

Our reformulation applies R_y rotations by reshaping to $(B, \text{left}, 2, \text{right})$ and precomputes the CNOT chain and R_z layer as a single diagonal phase vector over all 2^n basis states. The entangling layer then costs a single element-wise multiply.

We note that a closed-form product over qubits is *not* available here: the shared CNOT chain correlates the R_z phases, so the overlap does not factorise. This was verified empirically: with R_z disabled, a product form matches exactly; with distinct R_z parameters, it diverges. The gain arises from removing overhead, not from approximation.

This is a quantum-versus-quantum speedup. Simulating a circuit is necessarily more costly than the classical operation it parallels, and we measure the quantum branch at 5–14× slower than the classical path at every qubit count from 2 to 12.

13. Discussion

13.1. What the result does and does not show

We tested nine formulations; we do not claim no quantum-inspired method can help on this task, only that these nine, covering the three natural roles (similarity, feature extraction, inductive bias), do not. The structural account in Section 10 suggests that the issue lies in the data representation: real-valued embeddings from a face recognition backbone do not carry the structure that quantum-inspired methods are designed to exploit.

A quantum method might succeed if applied to raw pixel data (where entanglement could capture spatial correlations), or if the embedding space were redesigned to carry phase information. We regard both as speculative.

13.2. On the importance of protocol

The identity leakage documented in Section 4 means that prior claims of quantum advantage on kinship data are uninterpretable: a model achieving high accuracy under a leaked protocol may have learned to exploit the leak rather than the quantum feature. The corrected protocol proposed here is necessary for any future positive claim to be credible.

13.3. The purity regulariser as a methodological warning

Formulation 7 illustrates a failure mode in single-seed evaluation. A single run showed +1.3 ROC-AUC, a result consistent with the hypothesis that encouraging pure quantum states aids representation learning. Nine paired runs reversed the sign (-0.9, $p = 0.047$). Single-seed evaluation with quantum-inspired methods is particularly hazardous because the methods introduce additional stochasticity through their parameterisation.

14. Limitations

Single backbone. All results use frozen FaceNet embeddings. Whether the failures interact with backbone choice is untested; a backbone that produces embeddings with exploitable structure might yield different conclusions.

Small test sets. KinFaceW-I and KinFaceW-II yield 212 and 400 test pairs per fold, giving confidence intervals of roughly ± 2 points.

Classical simulation. All quantum computations are classically simulated. We cannot rule out that execution on quantum hardware might produce different results, though the mathematical operations are identical.

Fixed qubit count. We tested 2–12 qubits. It is possible that a much larger qubit count would exhibit different behaviour, but the computational cost of simulation grows exponentially.

15. Conclusion

We have evaluated ten quantum-inspired formulations for facial kinship verification, each against a capacity-matched classical control under a leakage-free grouped k-fold protocol. Nine do not improve on their controls; four are significantly worse. Their failures are structural: interference-based methods collapse because ℓ_2-normalised face embeddings carry no phase structure, and density-matrix methods fail because the number of images per identity is far too small to estimate a high-dimensional operator.

The ten formulations do not, however, form a uniform negative result. All nine that fail read their decision out of the quantum state as a single scalar. The tenth, evaluated under two readouts differing in nothing else, is indistinguishable from its control when read as a scalar (-0.012, $p = 0.26$) and exceeds it by $+0.132$ ROC-AUC when read as a per-qubit expectation vector ($p = 2.4 \times 10^{-7}$, robust across four seeds, five circuit configurations, and a shuffled-label ablation), and classical controls matched on readout width do not close the gap. Scalar readout is therefore a major limiting factor on this task — a distinction that a study reporting only scalar-readout formulations would have missed, and one that applies to any evaluation compressing a quantum state to a single number before the decision.

We stress what this does not show. Formulation 10 beats its capacity-matched local control but remains well below the global task baseline (0.674 against 0.8497 mean ROC-AUC) at 73.5× the inference cost, and classical simulation of a circuit is strictly more expensive than the classical path it replaces. These are different comparisons answering different questions, and only the second speaks to practical deployment. The result is a correction to how such components are evaluated, not a demonstration of quantum advantage.

Independently, we contribute a 380× GPU speedup for the simulated SWAP test, which may be useful in other quantum-simulation contexts. We also quantify the complete identity leakage in the evaluation protocol under which prior positive claims were made, and release a corrected protocol.

All code, data splits, evaluation scripts, and result artefacts are released at `https://github.com/svaibhav-7/Quantum_kinship_verification` so that every claim can be independently verified.

Reproducibility

Every numeric claim derives from JSON artefacts in `results/honest/`. The protocol is implemented in `src/splits.py`, `src/kfold.py`, and `src/ts_pairs.py`; results are regenerated by `scripts/evaluation/run_kfold.py`. The fast SWAP-test reformulation is in `src/quantum_fast.py`, validated against the reference simulator in `src/quantum_core.py` by 61 tests including gradient checks.

CRediT authorship contribution statement

Sasi Vaibhav: Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualisation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code, data processing scripts, and evaluation artefacts are publicly available at `https://github.com/svaibhav-7/Quantum_kinship_verification`. The underlying face datasets (FIW, KinFaceW-I/II, TSKinFace) are available from their respective authors.

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